
Pattern Arithmetic: Higher Truths from Assembly

Sandeep Lakshminarayan Chiluveru

October 7, 2026

Abstract

This is a second companion paper to *Pattern Arithmetic: Typed Algebra of Bounded Units on a Phase-Colored Sphere*, alongside *Pattern Arithmetic: Inference from Bowls and Paths*. The first defines assembly ($\bowtie$) and an enclosure in which units may descend an energy and rest as one camp or several; the second reads a bowl of evidence into a verdict, but never asks what a settled camp itself is entitled to be. This paper collects what is currently known about that question, and it is deliberately small. Two points are already closed, as direct consequences of choices the companion paper already made rather than of anything built here: the actor/patient question, settled at the level of a label rather than a difference in motion, and the absence of astrophysical mass segregation, a forced consequence of excluding volume from motion. One point is genuinely open — fine-grained multi-vector matching, where an atom’s own informational vectors, not its coarse derived vector, decide how it bonds. What this paper does not yet do is the harder thing its own title promises: say what a frozen camp, read and re-read at successive checkpoints, is entitled to be called, and whether that naming is itself a higher unit of truth or only a description of one. That is stated here as this paper’s own direction, not answered as its content.

1 Introduction

A camp settles. Something froze at a resolution η , with a body $\vec{P}_c$ and a membership — and the companion paper is explicit that the body is a description, measured after the fact, never a destination the motion sought. Read again at a finer η , or later in time, the same units may freeze into a different shape, or one pile where there were two: a camp is a checkpoint, not a fixed thing, and two checkpoints close enough to each other are read as the same freeze only by a further, named allowance (companion paper, §9.3). None of

this says what a checkpoint *is*, in the sense a bowl of evidence becomes a verdict in the other companion paper. That is the question this paper exists to carry, separately from inference, so that it can grow on its own rather than as an appendix to a paper about something else.

What follows is not yet that answer. It is the ground already cleared: two questions the companion paper’s own axioms settle as a side effect, stated here plainly since neither was recorded at the time, and one question about how an atom’s inner structure governs assembly that remains genuinely open. The larger question — what a settled camp, read against known patterns or against its own history of checkpoints, is entitled to be named, and whether that naming closes into a symbol a higher unit could itself be built from — is named in the abstract and left there. Early exploration toward it (reading a settled camp against known references, and characterizing when such a reading is trustworthy rather than lucky) exists only as worked, unpublished calculation; none of it is claimed here as a result.

2 What Is Inherited

We use, without re-deriving, the following from the companion paper. Equation and axiom numbers refer to that paper.

- The atom $x = (r, \hat{n}, \chi, \tau) \in \mathcal{S}$, with $r \leq 1$ a normalized strength (Axiom 1).
- Assembly, $\bowtie$: two units related while both remain intact, $A \bowtie B = (A, B, \kappa_{AB}, U_{AB})$, with $U_{AB} < 0$ for nonzero strengths and never repelling (Axiom 6, §2.9).
- The enclosure: external seats $\vec{P}$ descend the energy $\sum U_{ij}$ without mass or momentum until a stated force threshold η is crossed; a configuration below it is a *checkpoint*, not an exact equilibrium, since the pull between any two units of nonzero strength never reaches zero at finite distance (§9.1–9.3); two checkpoints are read as the same freeze only under a named allowance ϵ_{allow} (§9.3).
- Informational vectors (Axiom 7, §2.10): an atom holds Q capped informational vectors in N cores, of which $D \leq N$ are distinct, and one uncapped derived sum V_{derived} . Coarse assembly runs on the derived pair; the atom’s own vectors are not consulted by U_{AB} unless named explicitly.
- Collapse, $\oplus$, and UnCollapse, $\ominus$ (§3.3–3.4): several atoms combined into one with every core kept, $N = \sum N_i$ and $Q = \sum Q_i$; not idempotent, $A \oplus A = A^{(2)}$; $\ominus$ returns the atom without one core.
- The atom as a data unit (§2.11): a bowl is the set, a Collapsed atom the counted bag, the path the ordered tape, and the cores of an atom a dictionary keyed by identity. Assembly relates atoms as such data units, and it orders them or counts them no further than the atom itself does.
- Meta vectors (§2.11): values that describe an atom beside its content and are no part of what it says — the seat $\vec{P}$ and V_{derived} among them, with a timestamp, an expiry, a location and a link. They are uncapped, counted in neither N nor Q , and changed by no verb; Assembly reads V_{derived} and the seat.

- *Levels are closed*: a camp carries no representation of what it constitutes, and the enclosure supplies no comparison by which a checkpoint could name itself.

3 What Assembly Already Settles

Two settled points are worth recording plainly, since both follow from Axiom 7’s exclusion of $\mathcal{I}_{\text{vol}}$ from motion (companion paper, §2.10 and §9.2) rather than from anything built here.

The actor/patient question. Closed at the level of a label: polarity ($|r_A - r_B|$) distinguishes a mutual bond from a one-sided one, and no further behavioral mechanism is missing — one was tested directly and found to require exactly the volume-weighted motion that exclusion rules out on stability grounds, so the honest final scope of this question is a description of the bond, not a difference in how it moves.

Mass segregation. The astrophysical phenomenon — stronger members sinking toward a group’s centre — was named early in the companion paper’s own development as an expected consequence, while its dynamics was still built around a mass-like quantity. Once motion settled as excluding $\mathcal{I}_{\text{vol}}$ entirely, that mechanism no longer applies here, confirmed directly: equal force with nothing dividing it gives identical movement regardless of strength or volume. The astrophysical phenomenon keeps its real name; nothing in this framework currently reproduces it, since the quantity that would drive it was deliberately kept out of motion. The companion paper now says the same of its own enclosure: no sorting of members by strength (§9.5).

4 Open: Fine-Grained Multi-Vector Matching

Coarse assembly runs on the derived vector (companion paper, Axiom 7), so U_{AB} never needed to generalize; simultaneous multi-partner bonding via informational vectors directly remains optional, with an unconstrained double sum and an exclusive (Hungarian) matching identified as the two candidates. They agree whenever alignment quality is roughly uniform across candidates, but rank-order pairing can be strictly worse than optimal when strength and alignment pull in different directions — verified with a constructed adversarial case (-1.17 for rank order versus -1.68 true optimum).

5 Scope and the Question This Paper Is For

This paper is small on purpose. It records what the companion paper’s own choices already settle about assembly, once, so that neither has to be rediscovered; and it names one open question about how an atom’s own informational vectors should govern multi-partner bonding, where the coarse derived-vector treatment is known to be incomplete.

What it does not do is name a checkpoint. A frozen camp has a body and a membership and, by the companion paper’s own accounting, nothing else — no shape, no representation of what produced it, no claim on what it should be called. Whether a checkpoint can be read against known patterns and trusted, whether a sequence of checkpoints under an allowance converges on something worth naming, and whether such

a naming is a genuine higher unit of truth or only a compact description of a lower one — these are the paper’s own stated direction, not its content. They are left here, deliberately, for the work this paper exists to carry.

Development Notes: Ideas in Progress

What follows is a working record, not a finished section of this paper. Each item is tagged by status — **[Settled]**, **[Verified]**, **[Proposed]**, or **[Open]** — so a later pass can tell at a glance what is actually established, what is argued but unbuilt, and what has been tried and genuinely not yet solved, without re-deriving that distinction from scratch. Nothing here is claimed as this paper’s content; it is the material its later sections would be built from, once built.

Structure and organization

[Proposed.] Assembly as a layer beneath Inference, not a peer to it: Inference reads whatever atoms already exist; Assembly asks what happens when atoms relate to each other before anything is read. A camp is not evidence, by the companion paper’s own words — its body is “measured afterwards. . . a description rather than a destination” — and this paper’s own abstract already names camp-naming as unanswered here, not in Inference. The layering is implicit in language already sitting in both documents; stating it as the organizing structure is what is new. Not yet written into the Introduction.

[Proposed.] An eight-part restructuring built around that layering: a reframed Introduction stating the one-directional dependency; *What a Camp Is*, an axiom-like core of invariant camp properties; *Two Modes — Free and Directed Assembly*; *Representing a Camp*; *Structural Diagnostics*; the existing inherited and open-question sections unchanged; a closing statement rewritten around the layering. Two of eight sections need no work; the other six, including the entire axiom-like core, are still only conversation.

[Proposed.] Assembly’s own closure axioms should carry their own numbering, not continue the companion paper’s — justified by that paper’s own Axiom 0 discipline: a snowflake’s native law is recorded as refused a map onto the sphere rather than forced into $SO(3)$, because a system of axioms built for one kind of question should not silently claim authority over a different kind by mere continuation of its own numbering. Closure is that kind of different question relative to the companion paper’s atom-level axioms.

[Settled.] The test for what actually earns axiom status here — forced (no freedom, like π), genuinely invented but low-stakes (a normalization, like $r \leq 1$), or a real structural choice (a different choice produces a different theory) — is already written into this paper, below, as the parked aside for the companion paper’s Thesis. Not repeated here.

[Proposed.] A first candidate that passes that test: what a freeze keeps and what it discards. The companion paper says a camp’s body is a description, not a destination, and that a camp is a checkpoint, not a destination, but never formalizes exactly what survives a freeze versus what does not. Not derivable from the companion paper’s own axioms; not yet drafted as a formal statement.

Representing a camp

[Verified.] Resultant versus dominant-vector representation, tested on identical seats and multi-vector atoms: 16.0% of contested pairs disagree (120/749 across 15 trials) on

who ends up grouped with whom, depending solely on which representation is used.

[**Verified.**] With strength made genuinely irrespective — removed not just from the pull’s magnitude but from which vectors get compared at all — the two candidates already named in this paper’s own open question behave very differently. The unconstrained double sum collapses to a single undifferentiated group in every trial tested, confirmed not a settling-parameter artifact by a same-population sanity check against the ordinary single-vector case. Hungarian best-match survives with real, differentiated structure (3–5 groups per trial, genuine size variation). Between this paper’s own two candidates, one of them is now shown not to work for this case.

[**Proposed, untested.**] Centrality, treating U_{AB} as a weighted graph and reading a camp’s most central member, as a fourth candidate. Not yet tested at all.

Structural diagnostics

[**Verified.**] Graph community detection is not a faster substitute for physical settling. Tested properly, with distance included in the edge weights after an initial pass without it was caught as an unfair comparison: 69.7% of contested pairs disagree, and community detection consistently finds fewer, larger groups than settling does. Diagnosis: settling captures the dynamic, cascading resolution of competing pulls over time; a one-shot graph read of the starting configuration cannot see that resolution happening.

[**Proposed, untested.**] Hierarchical clustering as the formal version of a sentence the companion paper already states in words: “a finer η runs longer and may see groups merge that a coarser one froze apart.” That is a dendrogram, described in prose, never built.

The query/selection border case

[**Verified.**] A query matching two named vectors while excluding a third produces a genuinely non-transitive relation — global transitivity checked directly at exactly zero, with individual components confirmed to be chains rather than cliques (one three-member component held only 2 of its 3 possible internal edges). This is not a camp in this paper’s sense; it is a sparse, bridge-like network, closer to what sociology calls weak ties or structural holes.

[**Flagged, unresolved.**] This looks structurally closer to Inference’s own open selection rule — who enters, under what condition — than to Assembly’s job of settling a membership once given. Which paper should own it is not decided.

Systems as atoms

[**Settled.**] Vocabulary: *camp seat* for a camp’s spatial displacement (extending the companion paper’s own existing “seat” register) versus *camp signature* for its effective informational reading (checked clean of collisions across all three documents). These were previously, silently, one conflated vector.

[**Verified.**] Never-repels survives the jump to the system level: checked directly on system-level pairs, minimum pairwise coefficient positive ($\approx 4.4 \times 10^{-7}$), nothing negative anywhere.

[**Verified.**] The clip-versus-bijection choice matters again at the system level, not only at the evidence level: the identical 16 camps produce 13 systems under the Hill

bijection (Inference paper, §2.1) and 11 under the naive clip, with different groupings, not merely different counts.

[Verified — the hypothesis’s central test.] Treating settled camps as new, system-level atoms produces genuinely different behavior from simply widening the grouping radius on the original atoms. Two-level construction: largest system holds 32% of all atoms (16/50). Naive coarsening of the same population at a comparable scale: 94% (47/50), a near-total collapse. Camps compressing into a signature before the next level’s dynamics run is not equivalent to loosening the same single-level process.

[Added in the sync to v15_28; for the author’s review.] The companion paper now has a verb for several atoms becoming one: Collapse, $\oplus$ (§3.3), which keeps every core, $N = \sum N_i$, and discards nothing. A camp read as a system-level atom would then be a Collapsed atom, and its V_{derived} the sum of its members’ sums. Whether that matches the camp signature used in the tests above is not checked here; those tests promoted camps by their own construction.

Stability at the system level — the open frontier

[Open, actively unsolved.] Pure radial attraction with nothing to counteract it trends toward full collapse — matching the companion paper’s own stated Enclosure behavior, “left longer, all would merge into one,” and matching why naive coarsening above jumps almost directly to one giant group.

Checked against real astrophysics rather than assumed: globular clusters do sometimes merge, but genuine mergers cluster around specific circumstances — early galactic epochs, dense environments, close tidal encounters — not a generic, ongoing process. Most clusters, once in stable orbits, persist for billions of years without merging. The stabilizing mechanism in every such case is orbital motion: angular momentum.

[Tension identified, not resolved.] Real angular momentum needs mass ($L = r \times mv$) and inertia, motion that persists between moments rather than being reset by the current force alone. This paper’s own motion is deliberately mass-free — the reason mass segregation does not reproduce here, already a settled point above — and deliberately overdamped, velocity set directly by force with no memory of the step before. A literal transplant of angular momentum would undo both of those deliberate choices at once.

[Tested, did not work; problem remains open.] A proxy attempt: a tangential force built from each atom’s own $\hat{n}$, already present and otherwise unused for force direction, added to the radial pull. The first check looked promising — the largest group shrank from 97.5% to 7.5% of the population as the tangential term strengthened. A second, independent check showed why that was misleading: the system was not settling at all at the stronger values, only still moving (last-step size ≈ 0.40 – 0.44 , against ≈ 0.0003 for the undisturbed case). An overdamped system has nothing to damp a tangential force the way real inertia damps orbital perturbation, so the raw proxy trades collapse for motion that never settles, rather than producing anything stable. The fix needs a genuine damping mechanism, not yet designed. This is the most current, least resolved thread in this record.

Filed from the shelf: destinations and status

Three destinations only: the unit paper (comments, never new axioms), Inference (saying: bowls, tapes), and this paper (sitting: shapes, naming later). A *comment* is one or two sentences sharpening a claim already in the unit paper; anything needing a new operator or a worked camp is not a comment.

[**Settled.**] The rule that sorts every new thought: presence is not order; sitting is not saying; a name is later than a camp; pictures borrowed, laws not. A sentence that wants one verb that sits, walks and names is not ready.

[**Proposed, Inference.**] The swap test stays the fork: order changes the verdict, walk; otherwise Mix. A paper tape or piano roll is a picture of T : holes are pins, the crank is cadence, two holes named alike are two visits. Both senses of the axis, T and T^{-1} , become play and rewind. One bowl may feed two walks, T_{poem} and T_{tune} : two tapes, not many cranks at once. Film: one atom per frame is a lexicon choice, $\uplus$ is the cast ($\oplus$ if repeated frames must be counted), the movie is the ordered visits, T_{24} is a metronome, and a projector walks the reel without Mixing it. A camp may suggest a tape; walking it is still Inference. Attractors used to call up, down, hold or the next frame are comparisons to pins after a walk or a Mix, with the right to abstain: a costume, not a market model. Still open there: comparison weights, the selection rule, and one clean sequential numerical example.

[**Flagged, undecided.**] The quantum SWAP test, the standard overlap-estimation circuit, is a different thing from Inference's swap test, and the shared name is the same trap as "collapse" and "singularity" were. A related-work sentence marking the boundary helps, but a renamed test (an order test, say) may serve better. Any citation added there needs to be checked, not recalled.

[**Proposed, this paper.**] Clouds, chains and candle-files are shapes of rest under one U ; a chain or procession is a skinny cloud. A neighbour graph ("next") is a data structure on seats after rest, not T . A query, policy or stimulus may reweight the pull or pick a subgraph; walking the subgraph is Inference. Solid, liquid and gas are a costume for how free the seats are, and multi-zoom is a costume for coarse against fine η , with the same verbs returning only after a clump is named as a pin. Two clouds do not cancel by sitting: sitting is still $\uplus$ ($\oplus$ where the count must be kept), and cancelling is Mix on a shared axis, or taking a member out with $\setminus$ (a core, with $\ominus$). There is no annihilation operator. Information against anti-information is opposite χ on the same pin, Mix to black, not two gases consuming each other; it is separate from the alignment-offset repulsion tested earlier, which acts on opposed $\hat{n}$, not opposed χ . A seat becomes a pin only if promoted, and whether it may be is this paper's own question. A second learning layer comes only after a name exists.

[**Reconciliation, against this record's tests.**] Two lines of the shelf were written against the base enclosure and should be read with the tests above. First, "no repulsion" is true of the enclosure, but the Bounded attractor tested here needs an added short-range repulsion at each pair's own scale, so it is an Assembly extension, not something the unit paper already says. Second, "no cycle attractor without a new term in U " understates it: first-order descent of any potential has no periodic orbits, since the potential strictly decreases along every non-resting path, so no term in U can give a Cyclic attractor. It needed a persistent velocity state with radial-only damping; the one non-potential alternative tried, a tangential force, did not settle. Separately, the systems-as-atoms test promoted camps mechanically to atoms; the entitlement to a name remains open.

[**Proposed.**] Order of writing. The unit paper: optionally a few sentences (the habit, pin against seat, same verbs after naming), with a new version only if those sentences are actually in the PDF, never for clouds. Inference next: tape, film, the camp suggesting a tape, the boundary sentence. This paper when ready: shapes, neighbour graph, freeze vocabulary, the promotion question.

Aside: Notes Parked Here for the Companion Paper

This belongs beside *Pattern Arithmetic*’s own Thesis, not here. Two of the three notes below have since migrated into that paper’s Introduction (v15_28, §1) — what arithmetic is *for*, and precedent — and are kept here for the record. The first, on what earns axiom status, has not, and is held here until it does.

Not every axiom is earned the same way. Some are forced: π (3.14159265...) and e (2.71828182...) carry no freedom at all, falling out of their own definitions identically regardless of who does the deriving — discovered, not invented, in the strongest sense available. Some are genuinely invented, the way physics’ free parameters are: the fine-structure constant, the particle masses, Planck’s constant’s numerical value — structure discovered, specific value measured, with no known reason it could not have come out otherwise. And some, like the companion paper’s own $r \leq 1$, are neither: a normalization, a choice of scale, cheap to invent because no structural weight rides on the specific number — rescale it and nothing about the theory’s shape changes.

The test that actually sorts these, worth stating plainly: not whether a rule was discovered or invented, but whether a different choice would have produced a different theory. π fails this test by having no alternative to compare against. $r \leq 1$ fails it by mattering only as a scale. Mix as phasor summation, or Bundle’s idempotence, pass it — a different choice there is a different algebra, not the same one in different clothes. This is the reasoning underneath the companion paper’s own line, that it “states its own rules as axioms rather than discovering them”; it is the same standard this paper’s own closure axioms, wherever they are eventually stated, should be held to.

A second note, on what arithmetic is for. [*Now in the companion paper’s §1, v15_28.*] Also belongs beside the companion paper’s Thesis — restating it from a different angle rather than adding a new claim to it. The companion paper’s own Thesis states what an arithmetic *is*, structurally: a unit of truth and a small set of rules by which such units combine, obeyed consistently. This restates the same thing from what it is *for*: not the numeric operations themselves, but the discipline of reaching a conclusion, a truth, or a reality by at least one method that is consistent, reliable, and repeatable — numbers being one way of doing that, never the only one. Structure and purpose, the same claim seen from two directions; both belong together wherever the Thesis is eventually restated.

A third note, on precedent. [*Now in the companion paper’s §1, v15_28.*] Also belongs beside the companion paper’s Thesis. Pattern Arithmetic is not a new technique — people have been using instances of it, unnamed, for a long time. Pāṇini’s grammar, the rule system generating the whole of Sanskrit’s valid word-forms from a fixed set of roots and operations, has already been called mathematics by some commentators, though only weakly — a genuine, historical instance of a domain supplying units and a small, consistent set of combination rules, exactly as the Thesis describes. Chemistry is a

second, richer instance, its own combination rule already the subject of this paper’s own testing above. A third, deliberately the weakest case: placing a chair at a table is also an instance, since it can be done in one particular way, repeatedly, arriving at the same result each time — the Thesis’s own bar for what counts, met even by something this trivial.

A fourth note, on what the unit paper may say. Anything headed there is a comment, not an axiom, operator or worked camp. Candidates, each a label on the existing brick:

- The habit is old (Pāṇini, chemistry, table and chair as the weakest): four to six lines after the statement that counting is one arithmetic. Kinship of habit, not identity with $\mathfrak{U}$.
- Pin is not seat: two lines where $\bowtie$ first needs something outside the 4-tuple. A pin is an address on $\mathcal{S}$; a seat is $\vec{P}$ on the tray.
- Levels stay closed until a camp is named: one line added to the existing “Levels are closed” paragraph, that the same verbs return only after promotion to a new 4-tuple.
- Ideas borrowed, statutes not: one line in the claim-scope subsection pointing at solid, liquid, gas and multi-zoom as pictures.
- Mix still has no inverse: already in the published version, not reopened.

Status at v15.28: the first item is in the companion paper’s §1; the borrowed-pictures point is partly in its §9.5; the pin-against-seat and level-closure comments are not yet there. Not even as a comment: T_{24} , film, neighbour graphs, anti-information clouds, attractors for trades, learning loops, the quantum swap test.